

Serendipity design details

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Abstract

Math and design details for my character Serendipity, a composite mind.

DISCLAIMER AND WARNING

This is not a scientific paper or engineering advice. This document is only intended as creative world-building and has not been peer-reviewed or approved for any practical application. Use of these ideas in real-world designs or other serious application is entirely at the reader's discretion and risk.

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1 Models in use

Models in use:

UEIA-convergent		Felicity	
Working memory	4	Working memory	7
Learning rate	0.05	Learning rate	0.2
Parameters	1152921504606846976	Parameters	2889057280000000000
Storage req.	1.845e19 bits	Storage req.	4.623e20 bits
Fliprate req.	3.168e24 bits/s	Fliprate req.	7.940e25 bits/s

UEIA-convergent-bridge		UEIA-Human bridge-convergent	
Parameters	4503599627370496	Parameters	69868591406062435
Storage req.	7.206e16 bits	Storage req.	1.1179e18 bits
Fliprate req.	1.238e22 bits/s	Fliprate req.	1.92e23 bits/s

Human-UEIA bridge synapse count is calculated in accordance with the 1/215 rule which says that the bridge network usually has around 215 times fewer synapses than the model it is to be used with. Since this one bridges two different their size is averaged first.

$$\frac{1152921504606846976 + 2889057280000000000}{2} \cdot \frac{1}{215} = 69868591406062435$$

All of these models are meant to be inferenced at 1000Hz.

All of them use 16-bit parameters.

$$69868591406062435 \cdot 16bits \approx 1.1179e18bits$$

Given the 2748 bit flips per 16-bit multiply-accumulate operation:

$$69868591406062435 \cdot 1000Hz \cdot 2748bits \approx 1.92e23bits/s$$

2 Composite mind structure

The composite mind is fully-connected meaning that every brain node has a direct connection to every other brain node via a bridge, and every bridge has a direct connection to every other bridge via a meta-bridge.

The number of brain nodes is 16, as such the number of bridges is:

$$N_B = \frac{16 \cdot (16 - 1)}{2} = 120$$

And the number of meta-bridges in that case is:

$$N_{MB} = \frac{120 \cdot (120 - 1)}{2} = 7140$$

There is only one human brain in the network, and the other 15 brain nodes are UEIA-convergent minds, as such all bridges from Human to UEIA will be the UEIA-Human bridges. There are 15 UEIA-Human bridges (N_{UHB}).

All meta-bridges that connect the UEIA-Human bridges are going to also be UEIA-Human bridges, so the number of UEIA-Human metabridges is:

$$N_{UHMB} = \frac{15 \cdot (15 - 1)}{2} = 105$$

The remaining ones can be UEIA bridges, so their number (bridges and meta-bridges) is:

$$N_{UB} = N_B - N_{UHB} = 120 - 15 = 105$$

$$N_{UMB} = N_{MB} - N_{UHMB} = 7140 - 105 = 7035$$

This gives the final composition:

Serendipity composite mind composition			
element type	model	count	num.parameters
brain node	Felicity (human)	1	28 890 572 800 000 000 000
brain node	UEIA-convergent	15	1 152 921 504 606 846 976
bridge	UEIA-Human-bridge-convergent	15	69 868 591 406 062 435
bridge	UEIA-convergent-bridge	105	4 503 599 627 370 496
metabridge	UEIA-Human-bridge-convergent	105	69 868 591 406 062 435
metabridge	UEIA-convergent-bridge	7035	4 503 599 627 370 496

brain nodes total : 16

bridges total : 120

metabridges total : 7140

parameters total: 55 072 444 555 909 715 560

$$\begin{aligned}
&1 \cdot 28890572800000000000 + \\
&15 \cdot 1152921504606846976 + \\
&15 \cdot 69868591406062435 + \\
&105 \cdot 4503599627370496 + \\
&105 \cdot 69868591406062435 + \\
&7035 \cdot 4398046511104 = \\
&55072444555909715560
\end{aligned}$$

3 computational requirements

The whole network is to be inferenced at 1000Hz.

All parameters are 16-bit.

A 16-bit multiply-accumulate operation requires 2748 bit flips.

Size of the model:

$$55072444555909715560 \cdot 16bit = 881159112894555448960bits \approx 8.812e20bits$$

Fliprate required to run the model:

$$55072444555909715560 \cdot 1000Hz \cdot 2748bits = 151339077639639898358880000bits/s \approx 1.5134e26bits/s$$

4 Misc. properties

Working memory is simply the sum of the working memories of all brain nodes:

$$15 \cdot 4 + 7 = 67$$

Learning rate gives the statistical probability of her succesfully retaining the ability to solve a test task on the learning rate portion of the idala screenings.

UEIA has learning rate of 0.05, meaning that correspondingly the probability of NOT learning how to complete the test task is 0.95. Felicity's learning rate is set to 0.2 so the corresponding probability of failure is 0.8.

Each mind independently rolls for the chance to remember how to complete the task so for serendipity as a whole to fail, every single mind independently has to fail as well. The probability of that is just the product of the individual minds' failure chances:

$$0.95^{15} \cdot 0.8 = 0.370633$$

And thus the chance of success is the oposite of that:

$$1 - 0.370633 = 0.629367 \approx 0.63$$

5 Summary

Serendipity composite mind composition			
element type	model	count	num.parameters
brain node	Felicity (human)	1	28 890 572 800 000 000 000
brain node	UEIA-convergent	15	1 152 921 504 606 846 976
bridge	UEIA-Human-bridge-convergent	15	69 868 591 406 062 435
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metabridge	UEIA-convergent-bridge	7035	4 503 599 627 370 496

brain nodes total : 16

bridges total : 120

metabridges total : 7140

parameters total: 55 072 444 555 909 715 560

inference rate : 1000Hz

data storage requirement : 8.812e20 bits

fliprate requirement : 1.514e26 bits/s

working memory capacity : 67

Learning rate : 63%